

The background features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern and dynamic visual effect.

Engineering Management

Inventory Management

What is Inventory

- ▶ A **physical resource** that a firm holds in stock with the intent of selling it or transforming it into a more valuable state.
- ▶ Stock of items kept to meet future demand
- ▶ Purpose of inventory management
 - ▶ How many units to order
 - ▶ When to order

Discussion Point

- ▶ **Pressures for Low Inventory??**
- ▶ **Pressures for High Inventory??**

Pressures for Low Inventory

- ▶ **Interest or Opportunity Cost**
- ▶ **Storage and Handling Costs**
- ▶ **Taxes, Insurance and Shrinkage**

Pressures for High Inventory

Customer Service

Ordering Cost

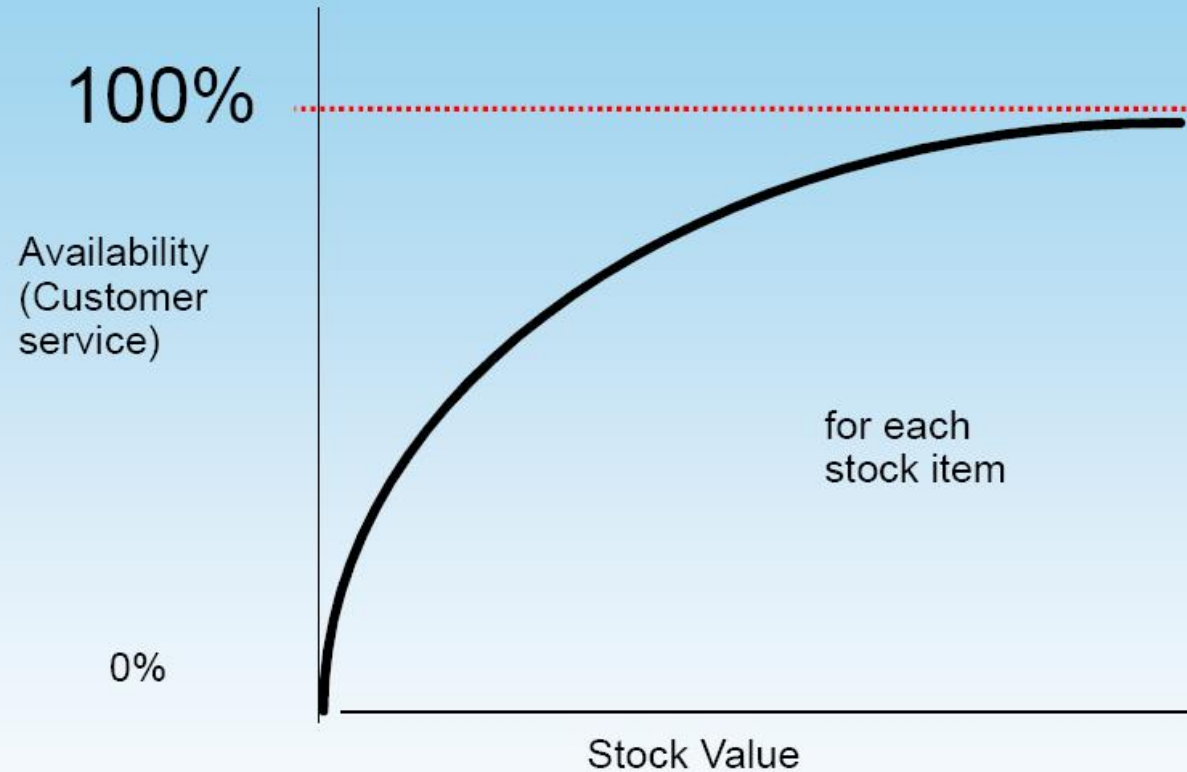
Setup Cost

Labor and Equipment Utilization

Transportation Cost

Payments to Suppliers

CUSTOMER SERVICE



Customer Service = $\frac{\text{Number of Orders Delivered On Time}}{\text{Total Number of Orders}}$

Types of Inventory

- ▶ **Cycle Inventory:** The proportion of total inventory varies directly with lot size is called CI.
 - ▶ Determine how frequently to order
 - ▶ What quantity??
 - ▶ Average Cycle Inventory= $(Q+0)/2=Q/2$

Types of Inventory

► **Safety Stock Inventory:**
**Order in advance...?? Lead time
Management**

Types of Inventory

- ▶ **Anticipation Inventory**
Seasonal..Incidental

Types of Inventory

- ▶ Pipeline Inventory is the average demand during the lead time.
- ▶ dL

Class Exercise

- ▶ Transcom Electronics, a manufacturer of consumer electronic goods, has five distribution centers (DCs) in different regions of the country. For one of its products, Air conditioner priced at \$350 per unit, the average weekly demand at each DC is 155 units. Average shipment size to each DC is 500 units and average lead time for delivery is two weeks. Each DC carries three week's supply as safety stock but holds no anticipation inventory.

Requirements:

- ▶ How much total inventory (cycle, safety and pipeline) does Transcom hold for all five DCs?

Inventory Reduction Tactics

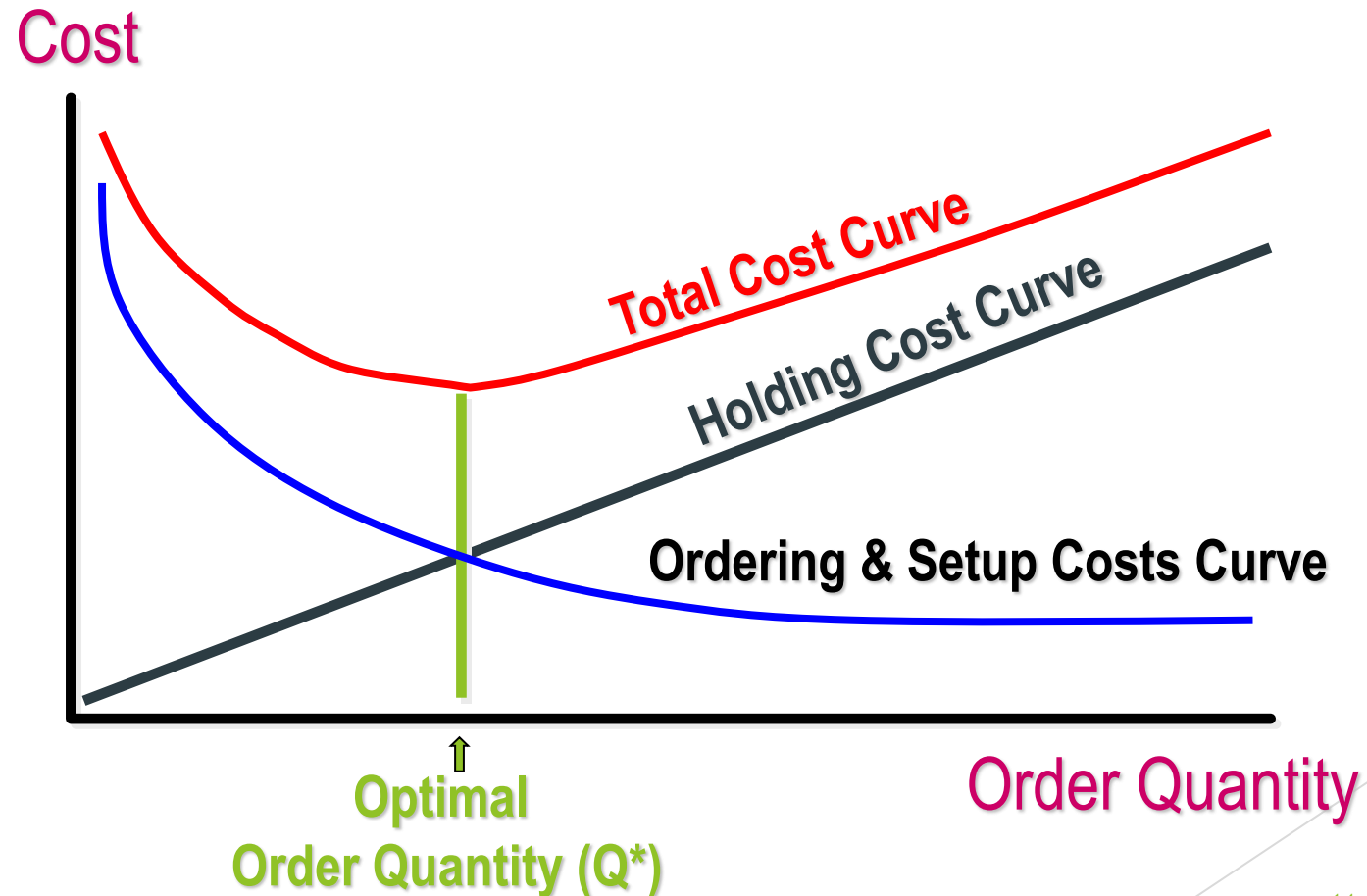
- ▶ **Cycle Inventory:**
 - ▶ **Reduce Lot Size**
 - ▶ But leads to increased Ordering Cost....
 - ▶ Streamline methods of ordering and changing setup.
 - ▶ Repeatability e.g. process design
- ▶ **Safety Stock:**
 - ▶ **JIT**
 - ▶ But leads to dissatisfied customers...
 - ▶ Better forecasting..reservation
 - ▶ Reduce Lead time such as local supplier
 - ▶ Reduce supply uncertainty
 - ▶ Equipments and labor buffer
- ▶ **Anticipation Inventory:**
 - ▶ Streamline demand and production rate
 - ▶ Portfolio diversification
 - ▶ Promotion
- ▶ **Pipeline Inventory:**
 - ▶ Lead time Mgt
 - ▶ Supply Mgt
 - ▶ Decrease Q

How???

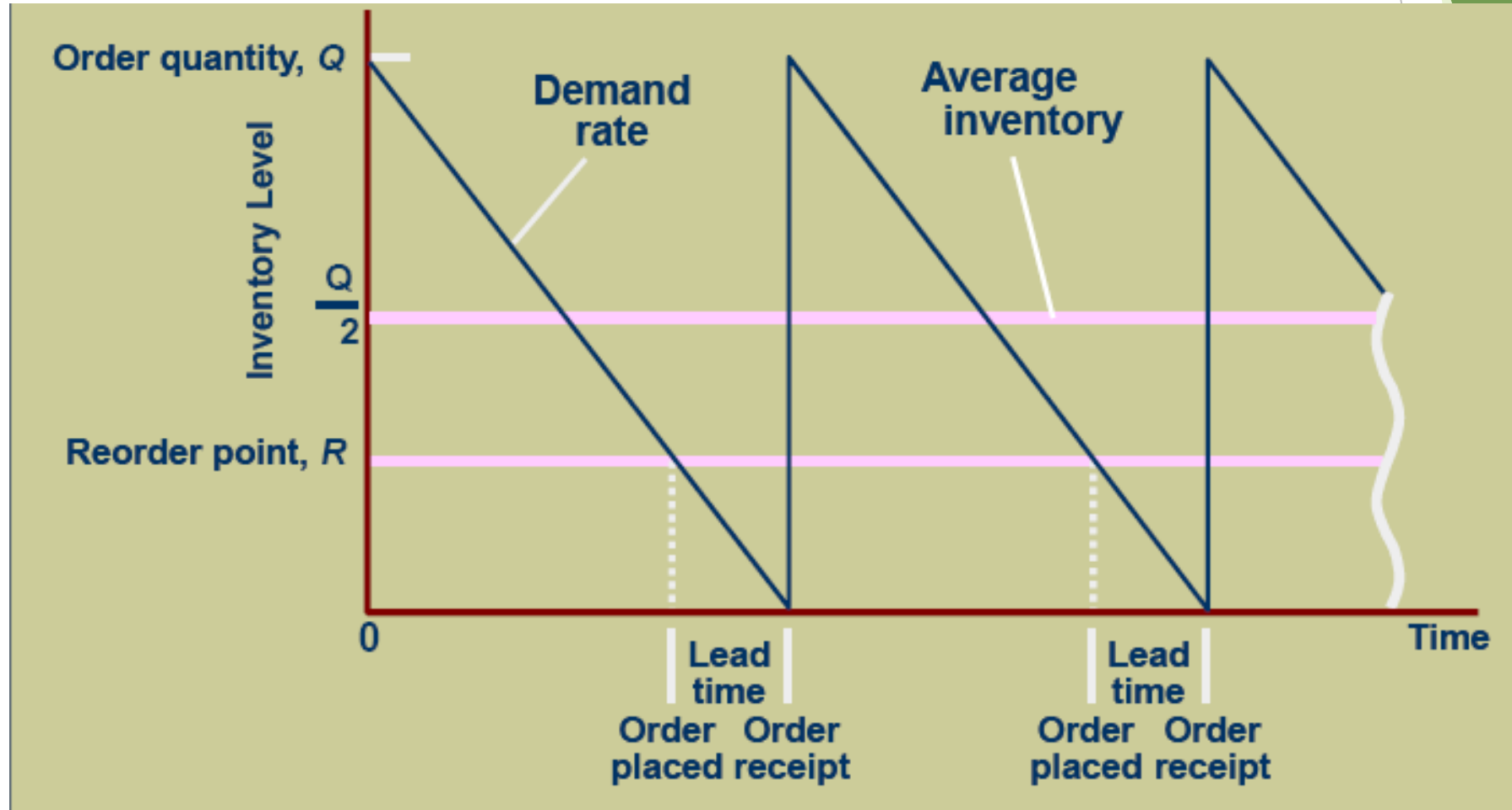
EOQ Model-Assumptions

- ▶ The demand rate is Constant.
- ▶ There are no other constraints on the size of each lot
- ▶ Two relevant costs are considered as inventory holding cost and fixed cost per lot for ordering or setup
- ▶ Decisions are made independently
- ▶ No uncertainty in lead time.

Combined Graph of AHC, AOC and TC



Inventory Order Cycle



Balance between two costs

- ▶ Annual Holding Cost=(Average cycle inventory)(Unit Holding Cost)
- ▶ Annual Ordering Cost=(No of orders/year)(Ordering or setup cost per order)

Total Cost= Annual Holding Cost+ Annual Ordering Cost

$$C=Q/2 (H)+D/Q (S)$$

C=total Annual Cost
Q=Lot size
H=Unit Holding Cost
D=Annual demand
S=Setup¹⁶ cost per order

Class exercise

One of the top-selling items in the container group at a museum's gift shop is a bird feeder. Sales are 18 units per week and the supplier charges Tk60 per unit. The cost of placing an order is tk 45. Annual holding cost is 25% of a feeder's value and the museum operates 52 weeks per year. Management chose a 390 unit lot size so that new orders could be placed less frequently.

What is the annual cost of the current lot size of 390 units? Would a lot size of 468 be better?

Answer

► $D = (18 \text{ units/week})(52 \text{ weeks/year}) = 936 \text{ units}$

► $H = 0.25(60/\text{units}) = \text{tk } 15$

Now,

LS=390;

$C = Q/2 (H) + D/Q (S)$

$= 390/2 (\text{tk } 15) + 936/390 (\text{tk } 45) = \text{tk } 3,033$

LS=75;

$TC = \{(75/2) \times 15\} + \{(936/75) \times 45\}$

$= 1,124.1$

When $Q=468$

$C = Q/2 (H) + D/Q (S)$

$= 468/2 (\text{tk } 15) + 936/468 (\text{tk } 45) = \text{tk } 3,600$

Derive the EOQ: Finding Q^* that Minimizes the Total Costs

Total inventory cost = Order (Setup) cost + Holding cost

$$TC = \frac{D}{Q}S + \frac{Q}{2}H$$

$$Q^* = \sqrt{\frac{2DS}{H}}$$

Classroom exercise

One of the top-selling items in the container group at a museum's gift shop is a bird feeder. Sales are 18 units per week and the supplier charges Tk60 per unit. The cost of placing an order is tk 45. Annual holding cost is 25% of a feeder's value and the museum operates 52 weeks per year. Management chose a 390 unit lot size so that new orders could be placed less frequently.

1. Identify EOQ.
2. Determine annual cost of the current lot size of 390 units?
3. Determine annual cost considering the EOQ?

Solution

Parameters:

$Q=390$

$D=936$

$S=45$

$H=15$

Annual Cost (when $Q=936$)

Order per year=2.4

Annual Ordering Cost=108

Annual Holding Cost=2,925

Annual Inventory Cost=3033

$EOQ=75$

Annual Cost (when $Q=75$)

Order per year=12.48

Annual Ordering Cost=561.80

Annual Holding Cost=562.50

Annual Inventory
Cost=1124.10

Classroom exercise

One of the top-selling items in the container group at a museum's gift shop is a bird feeder. Sales are 29 units per week and the supplier charges Tk60 per unit. The cost of placing an order is tk 65. Annual holding cost is 15% of a feeder's value and the museum operates 52 weeks per year. Management chose a 490 unit lot size so that new orders could be placed less frequently.

1. Identify EOQ.
2. Determine annual cost of the current lot size of 490 units?
3. Determine annual cost considering the EOQ?

Inventory Control System

- ▶ Continuous Review System
- ▶ Periodic Review System

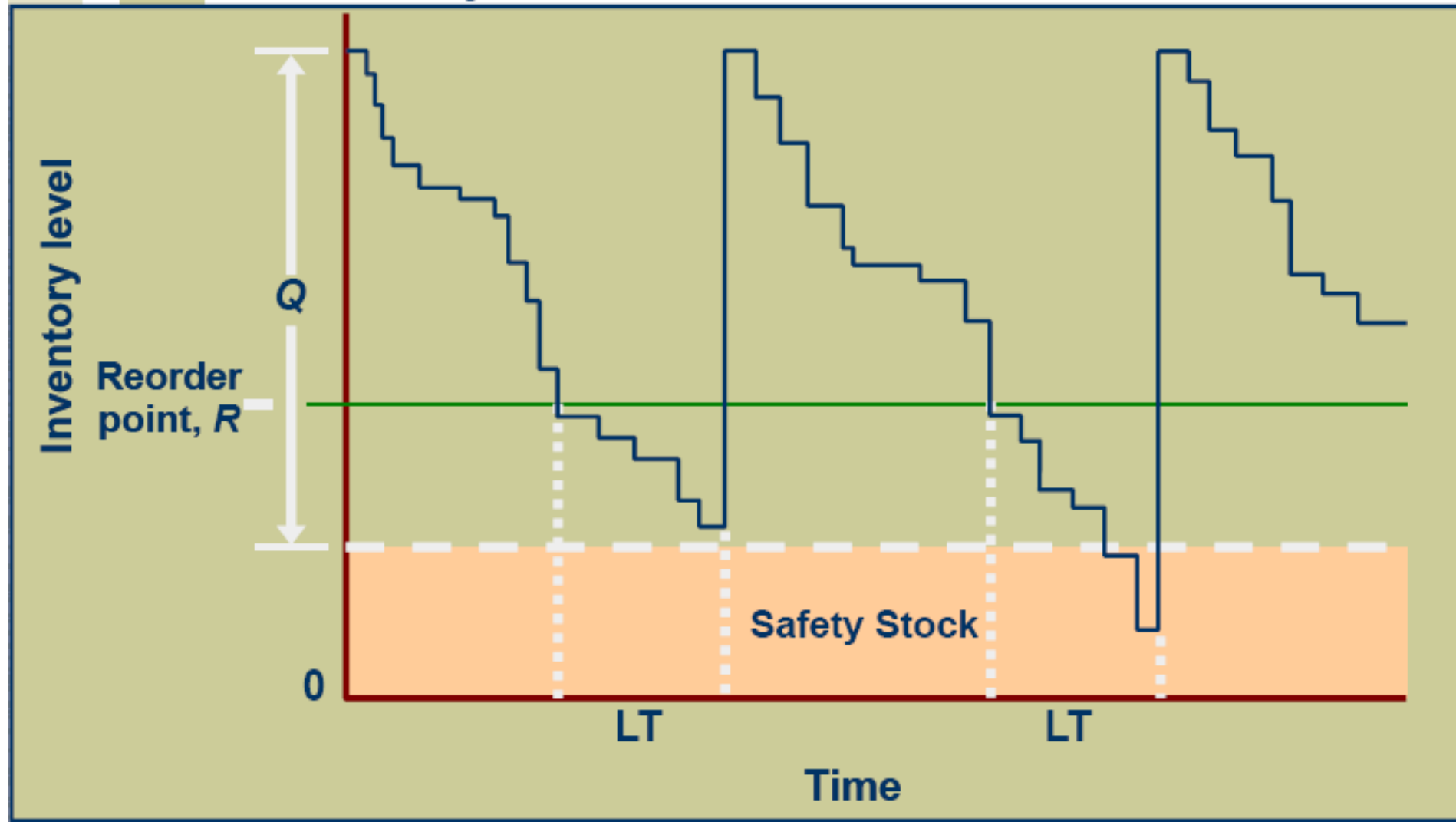
Continuous Review System (Q System)

- ▶ Remaining quantity of an item is reviewed each time a withdrawal is made from inventory to determine whether it is time to reorder. . In that case Q will be fixed but P is vary .
- ▶ The Inventory position (IP) measures the item's ability to satisfy future demand, relying only on SR and on hand inventory.
- ▶ Thus, $IP = OH + SR - BO$

Periodic Review System (P System)

- ▶ An item's inventory position is reviewed periodically rather than continuously.
- ▶ Thus review of item's inventory position IP every P time periods. Place an order equal to $(T - IP)$ where T is the target inventory. In that case P is fixed but Q will vary.

Reorder Point with a Safety Stock



END OF THE CHAPTER...